

**THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY**

**FUNDAMENTALS OF FLUID MECHANICS
MODEL QUESTION PAPER**

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark

(9 x 1 = 9 Marks)

1	Define the term Density.	M 1.02	R
2	Pressure below atmosphere is called.....	M 1.04	U
3	When Reynold's number is less than 2000, the fluid flow is generally	M 2.01	U
4	Venturimeter is used for measuring.....	M2.03	A
5	The ratio of Actual discharge to theoretical discharge is called	M 2.04	A
6	Name of valve used in the suction pipe of a centrifugal pump.	M 3.02	A
7	The in the reciprocating pump is the measure of the difference between theoretical discharge and actual discharge	M 3.06	U
8	To prevent the backflow of oil in to reservoir when the pump is not in operation is called valve	M 4.02	U
9	What is the function of air lubricator in pneumatic system?	M 4.05	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	Define the following (a) Specific Gravity (b) Kinematic viscosity	M 1.01	R
2	The pressure intensity at a point in a fluid is given as 35kPa. Find the corresponding height of fluid when the fluid is (a) Water and (b) Oil of specific gravity 0.8	M 1.03	A
3	State Bernouli's theorem and write the equation	M 2.01	R

4	Differentiate between steady flow and unsteady flow	M 2.01	R
5	A rectangular notch 1.8 m wide has a constant head of 4,6 m. Taking the coefficient of discharge as 0.62, calculate the discharge	M 2.04	A
6	List the major and minor head losses in pipes	M 2.05	R
7	What is meant by cavitation in centrifugal pumps	M 3.03	U
8	What is the difference between manometric efficiency and overall efficiency of reciprocating pump	M 3.04	R
9	Draw the layout of pneumatic system and mark the basic elements	M 4.05	R
10	What is the principle of a hydro- pneumatic system	M 4.05	R

PART C

III Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

1	What is the difference between U-tube differential Manometer and Inverted U -tube manometer? Explain the difference with suitable sketches. OR	M 1.05	U
2	A U-tube differential mercury manometer is connected to two pipes A&B. Pipe B is 60 mm below pipe A. The specific gravity of liquid in pipe A & B is 1.6 & 0.85 respectively. Mercury level in the left limb is 80mm below the center of pipe A. Find the pressure difference between two pipes in kPa if the level difference of mercury in the two limbs of the manometer is 120mm.	M1.05	A
3	A horizontal venturimeter is provided in a pipeline 300mm diameter carrying oil of Sp. Gravity 0.8. The throat diameter is 150mm.If the differential manometer gives a reading of 250mm of mercury find the discharge in liters per second. Assume coefficient of discharge as 0.97 OR	M 2.03	A
4	State and explain Bernouli's theorem	M 2.02	U
5	A double acting reciprocating pump has a stroke 300 mm, piston diameter 150mm, delivery head 20m and suction head 3m. If the pump is working at 60 rpm, Find the actual and theoretical power required to drive the pump. Assume the efficiency as 75%. OR	M 3.04	A
6	Explain with neat sketch the working of a centrifugal pump	M 3.02	U
7	Explain the basic components of a hydraulic system with a neat layout. OR	M 4.02	U

8	Describe the working of an air filter used in pneumatic system	M 4.03	U
9	A simple manometer is used to measure the pressure of oil of specific gravity 0.8 flowing in a pipe line. Its right limb is open to the atmosphere and left limb is connected to the pipe. The center of the pipe is 90mm below the level of mercury in the right limb. If the difference of mercury level in the two limbs is 150mm, determine the absolute pressure in pipe in MPa. Take the specific gravity of mercury as 13.6. OR	M 1.05	A
10	A pipe of 500m long is conveying water with a velocity of 10m/s. Find the suitable diameter of the pipe if the head loss due to friction is 3.4 m. Take $f=0.001$	M2.06	A
11	Explain different efficiencies of a centrifugal pump OR	M 3.04	U
12	Explain the working of a solenoid operated valve	M 4.03	U